UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education Ordinary Level

CHEMISTRY **5070/01**

Paper 1 Multiple Choice **October/November 2007**

1 hour

Additional Materials: Multiple Choice Answer Sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

* 1 6 7 8 6 8 9 0 2 1 *

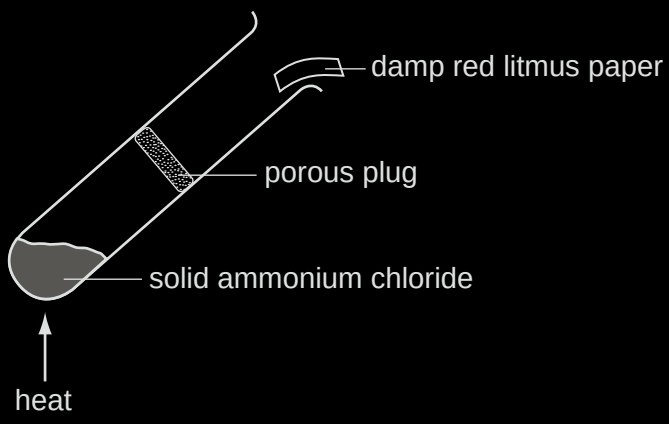
READ THESE INSTRUCTIONS FIRST

Write in soft pencil.
Do not use staples, paper clips, highlighters, glue or correction fluid.
Write your name, Centre number and candidate number on the Answer Sheet in the spaces provided unless this has been done for you.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.
A copy of the Periodic Table is printed on page 16.



Which change occurs to the damp red litmus paper in the experiment above?

A remains red

B turns blue and is then bleached

C turns blue and remains blue

D turns blue and then turns red

3 Compound **X** reacts with some metals to liberate hydrogen and is used to make fertilisers.

It gives a white precipitate when added to aqueous barium nitrate.

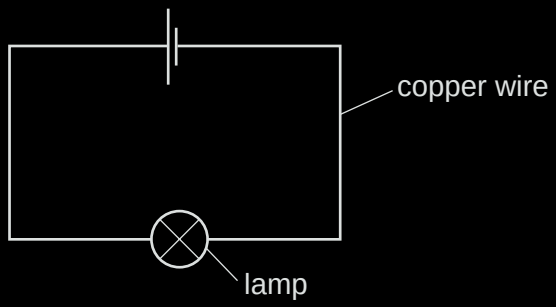
What is **X**?

A ammonium sulphate

B hydrochloric acid

C potassium nitrate

D sulphuric acid



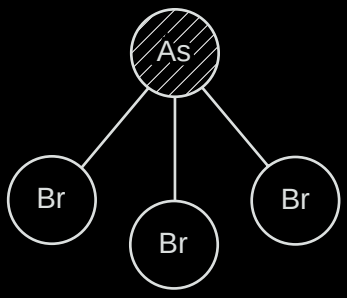
Which process takes place in the copper wire?

A Electrons move along the wire to the negative terminal, positive ions stay in position.

B Electrons move along the wire to the positive terminal, positive ions move to the negative terminal.

C Electrons move along the wire to the positive terminal, positive ions stay in position.

D Negative ions move along the wire to the positive terminal, positive ions move to the negative terminal.



Which properties could be correct for arsenic bromide?

	melting point / °C	electrical conductivity at room temperature
A	28	does not conduct
B	39	conducts
C	650	conducts
D	755	does not conduct

12 The equation represents the action of dilute nitric acid on copper.

$$x\text{Cu} + y\text{HNO}_3 \rightarrow x\text{Cu(NO}_3)_2 + 4\text{H}_2\text{O} + 2\text{NO}$$

What are the values of x and y?

A $x = 1$, $y = 4$

B $x = 1$, $y = 8$

C $x = 3$, $y = 4$

D $x = 3$, $y = 8$

13 Which statement about the substance formed when a given mass of an element burns in excess oxygen is **always** correct?

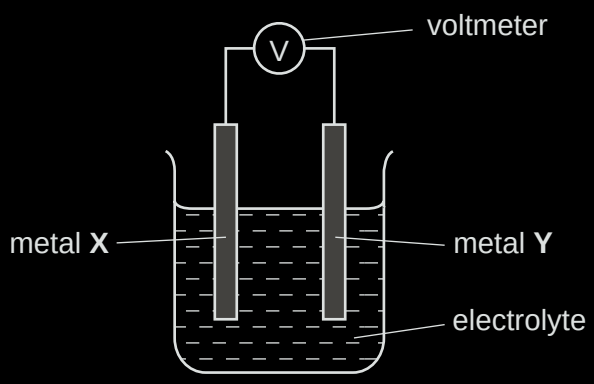
The substance formed is

A denser than the element.

B greater in mass than the element.

C soluble in water.

D white in colour.



Which two metals produce the highest reading on the voltmeter?

	X	**Y**
A	magnesium	copper
B	magnesium	iron
C	zinc	copper
D	zinc	iron

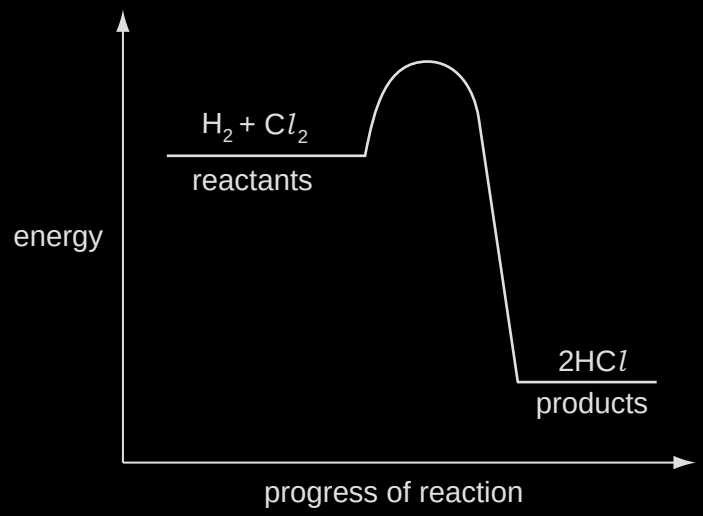
16 In which process is energy released?

A electrolysis of water to form hydrogen and oxygen

B forming a hydrogen molecule from two hydrogen atoms

C fractional distillation of crude oil

D photosynthesis



What information about this reaction does the diagram show?

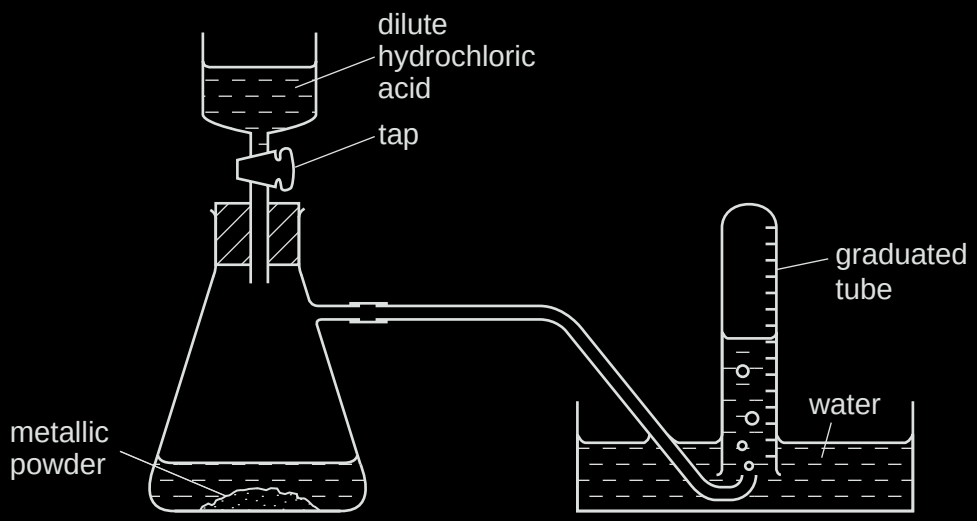
	type of reaction	sign of enthalpy change, ΔH
A	endothermic	negative
B	endothermic	positive
C	exothermic	negative
D	exothermic	positive

18 Carbon dioxide was produced when a given mass of zinc carbonate reacted with excess hydrochloric acid.

Which result shows what would happen if the reaction were repeated at a higher temperature?

	volume of carbon dioxide	rate of reaction
A	same	faster
B	same	slower
C	greater	same
D	greater	faster



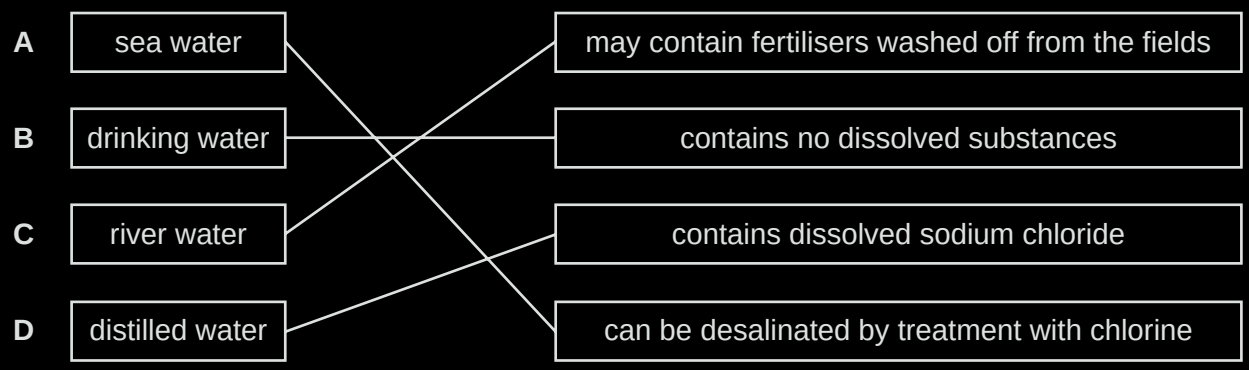


The experiment is carried out three times, using the same mass of powder each time but with different powders:

- pure magnesium
- pure zinc
- a mixture of magnesium and zinc

Which powder gives the greatest volume of hydrogen and which the least volume?

	greatest volume of H_2	least volume of H_2
A	magnesium	zinc
B	magnesium	the mixture
C	zinc	magnesium
D	zinc	the mixture



35 When cracked, one mole of a compound **X** produces one mole of propene and one mole of hydrogen.

$$\mathbf{X} \rightarrow C_3H_6 + H_2$$

What type of compound is **X**?

A an alcohol

B an alkane

C an alkene

D a carboxylic acid

36 When ethanol is left standing in the air for some time it becomes acidic.

Which equation represents this change?

A $CH_3CH_2OH + CO \rightarrow CH_3CH_2CO_2H$

B $CH_3CH_2OH + O_2 \rightarrow CH_3CO_2H + H_2O$

C $CH_3CH_2OH + 3O_2 \rightarrow 2CO_2 + 3H_2O$

D $2CH_3CH_2OH + O_2 \rightarrow 2CH_3CO_2H + 2H_2$

37 A 10 cm^3 sample of a gaseous hydrocarbon is completely burnt in oxygen. The total volume of the products is 70 cm^3.

Which equation represents the combustion of the hydrocarbon?

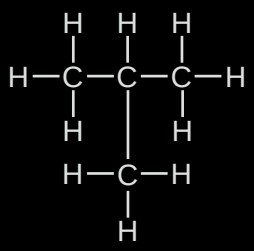
A $CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(g)$

B $C_2H_4(g) + 3O_2(g) \rightarrow 2CO_2(g) + 2H_2O(g)$

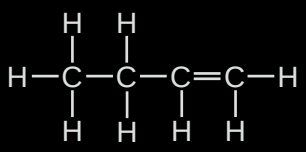
C $C_3H_8(g) + 5O_2(g) \rightarrow 3CO_2(g) + 4H_2O(g)$

D $2C_2H_6(g) + 7O_2(g) \rightarrow 4CO_2(g) + 6H_2O(g)$

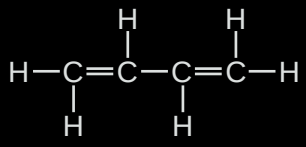
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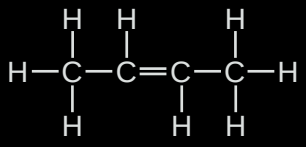
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3



4



Which structures are isomeric butenes?

A 1 and 2 **B** 2 and 3 **C** 3 and 4 **D** 2 and 4

DATA SHEET
The Periodic Table of the Elements

Group																
I	II										III	IV	V	VI	VII	0
							1 H Hydrogen 1									4 He Helium 2
7 Li Lithium 3	9 Be Beryllium 4										11 B Boron 5	12 C Carbon 6	14 N Nitrogen 7	16 O Oxygen 8	19 F Fluorine 9	20 Ne Neon 10
23 Na Sodium 11	24 Mg Magnesium 12										27 Al Aluminium 13	28 Si Silicon 14	31 P Phosphorus 15	32 S Sulphur 16	35.5 Cl Chlorine 17	40 Ar Argon 18
39 K Potassium 19	40 Ca Calcium 20	45 Sc Scandium 21	48 Ti Titanium 22	51 V Vanadium 23	52 Cr Chromium 24	55 Mn Manganese 25	56 Fe Iron 26	59 Co Cobalt 27	59 Ni Nickel 28	64 Cu Copper 29	65 Zn Zinc 30	70 Ga Gallium 31	73 Ge Germanium 32	75 As Arsenic 33	79 Se Selenium 34	80 Br Bromine 35	84 Kr Krypton 36
85 Rb Rubidium 37	88 Sr Strontium 38	89 Y Yttrium 39	91 Zr Zirconium 40	93 Nb Niobium 41	96 Mo Molybdenum 42	Tc Technetium 43	101 Ru Ruthenium 44	103 Rh Rhodium 45	106 Pd Palladium 46	108 Ag Silver 47	112 Cd Cadmium 48	115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	128 Te Tellurium 52	127 I Iodine 53	131 Xe Xenon 54
133 Cs Caesium 55	137 Ba Barium 56	139 La Lanthanum 57 *	178 Hf Hafnium 72	181 Ta Tantalum 73	184 W Tungsten 74	186 Re Rhenium 75	190 Os Osmium 76	192 Ir Iridium 77	195 Pt Platinum 78	197 Au Gold 79	201 Hg Mercury 80	204 Tl Thallium 81	207 Pb Lead 82	209 Bi Bismuth 83	Po Polonium 84	At Astatine 85	Rn Radon 86
Fr Francium 87	226 Ra Radium 88	227 Ac Actinium 89 †														

140 Ce Cerium 58	141 Pr Praseodymium 59	144 Nd Neodymium 60	Pm Promethium 61	150 Sm Samarium 62	152 Eu Europium 63	157 Gd Gadolinium 64	159 Tb Terbium 65	162 Dy Dysprosium 66	165 Ho Holmium 67	167 Er Erbium 68	169 Tm Thulium 69	173 Yb Ytterbium 70	175 Lu Lutetium 71
232 Th Thorium 90	Pa Protactinium 91	238 U Uranium 92	Np Neptunium 93	Pu Plutonium 94	Am Americium 95	Cm Curium 96	Bk Berkelium 97	Cf Californium 98	Es Einsteinium 99	Fm Fermium 100	Md Mendelevium 101	No Nobelium 102	Lr Lawrencium 103

*58-71 Lanthanoid series
†90-103 Actinoid series

Key

a
X
b

a = relative atomic mass
X = atomic symbol
b = proton (atomic) number

The volume of one mole of any gas is 24 dm^3 at room temperature and pressure (r.t.p.).